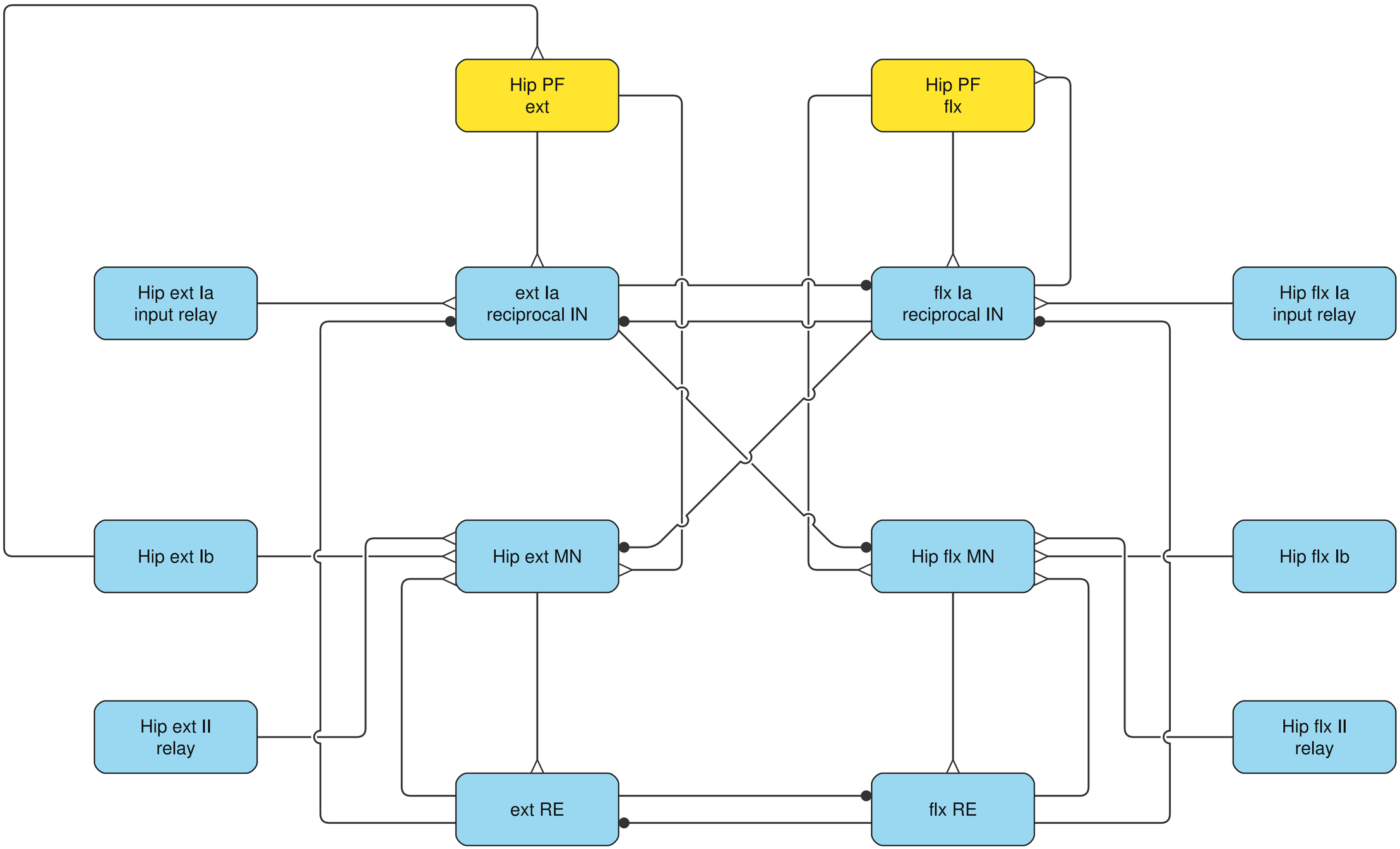
1 · Paired MN circuit: the saved W2L108 hip subnetwork



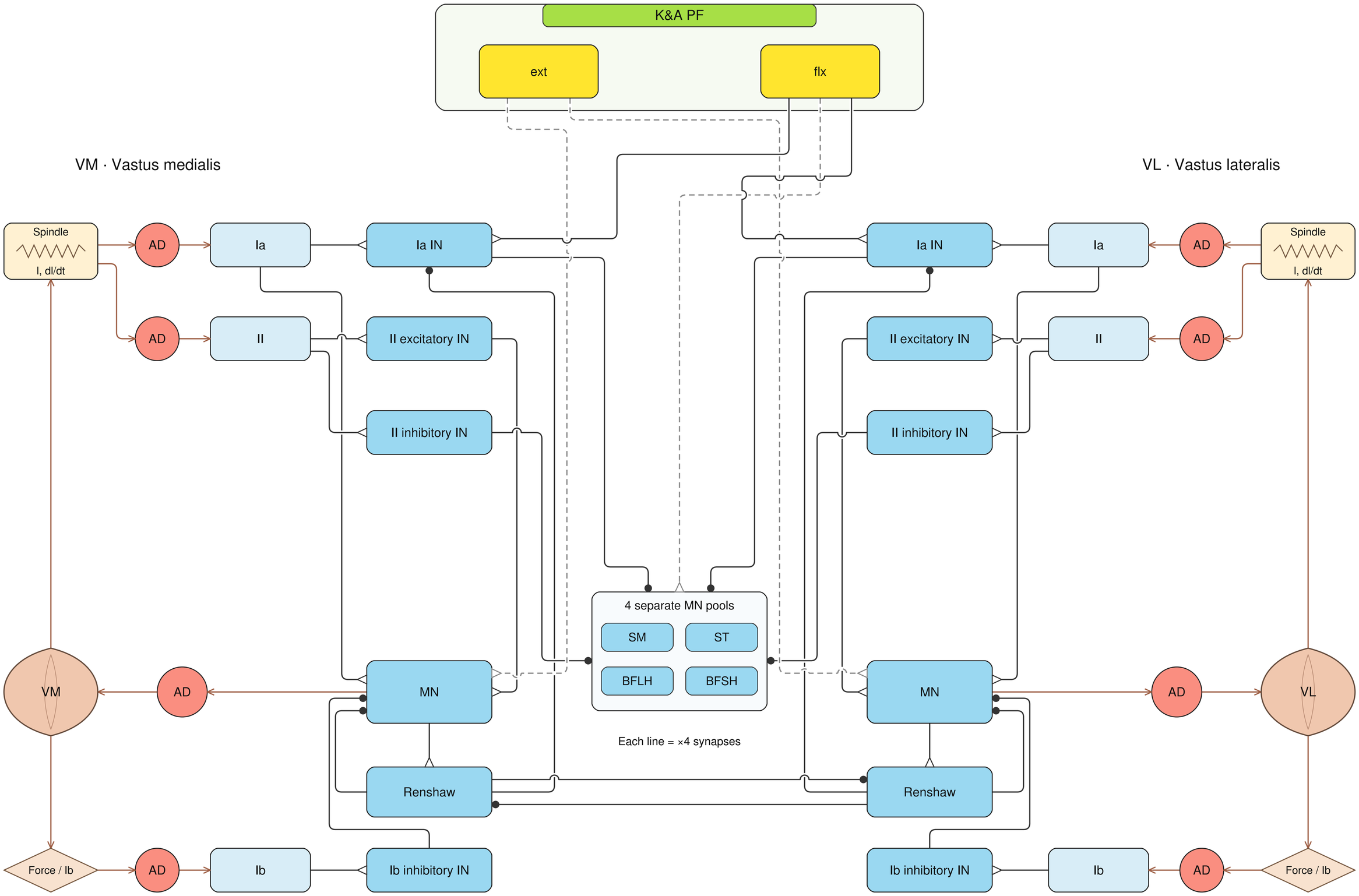
One extensor MN and one flexor MN; all saved neural connections among the displayed cells are shown.

Native naming: the peripheral “Ia IN” relays feed the reciprocal-inhibition cells named “Ia”.

Source detail: RE → own MN is excitatory here. RE → Ia and RE → opposite RE are inhibitory.

Open triangle: excitation (base on target). Filled circle: inhibition. Semicircular bridges: no junction.

Shared PF source · two muscle-specific reflex and physical loops



AD = map / adapter. Brown arrows: physical signals. Open triangles: excitation. Filled circles: inhibition.

Spindle and tendon symbols show encoded signals; muscle→spindle→AD→Ia/II and muscle→force sensor→AD→Ib are explicit.

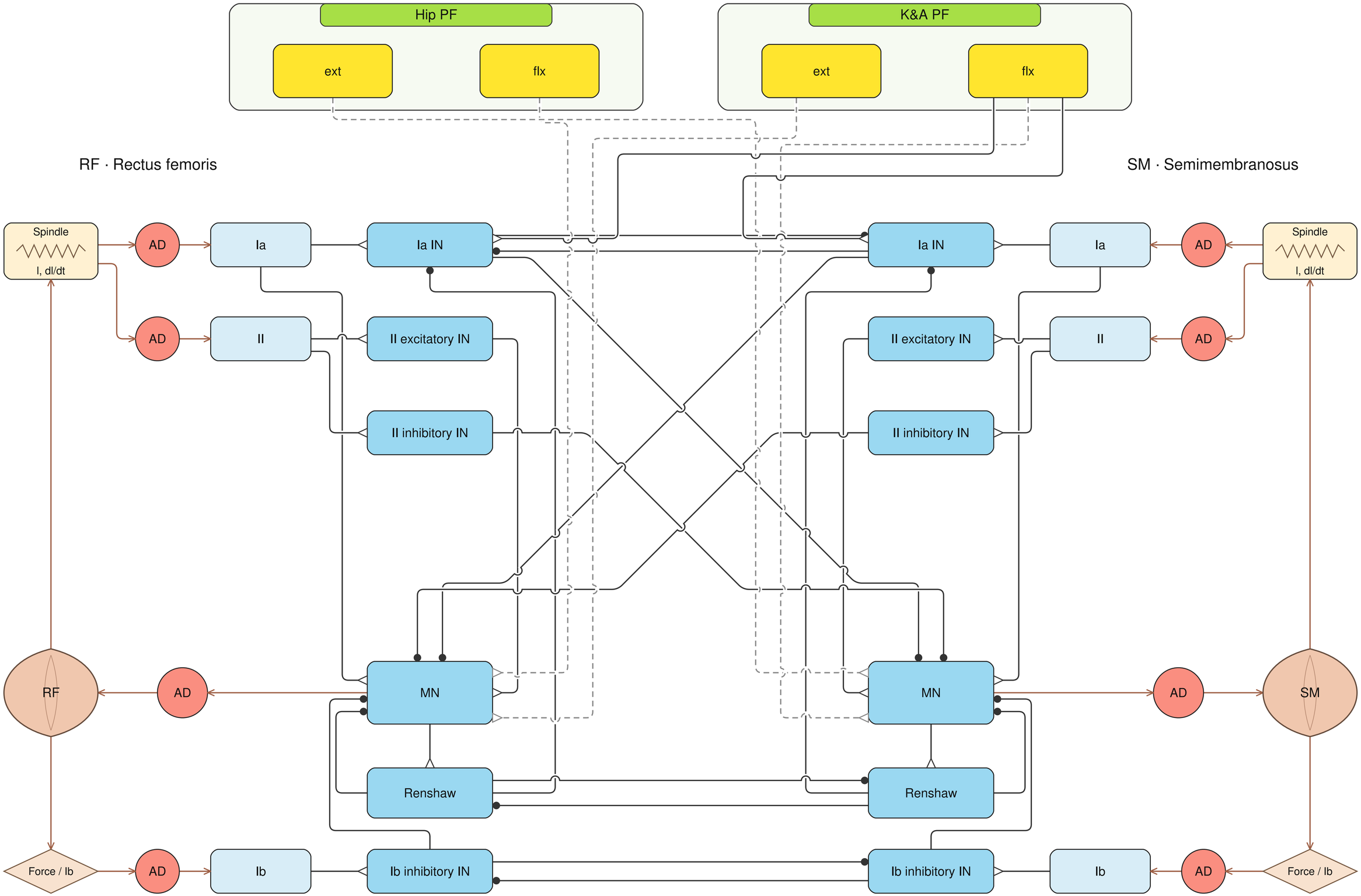
MN→AD→muscle carries activation. Each muscle retains its own MN, Ia, II, Ib, interneurons, and Renshaw cell.

Gray dashed PF→MN routes have effective g = 0. Central afferent→RG/PF projections are detailed in the preceding RG/PF sheets.

VM and VL share PF drive, but retain separate loops. The four antagonist destinations are grouped; every line to that group represents ×4 synapses.

SM = semimembranosus; ST = semitendinosus; BFLH/BFSH = biceps femoris long/short head. Their own local loops are outside this crop.

Biarticular muscles · PF inputs, local reflexes, and physical maps



AD = map / adapter. Brown arrows: physical signals. Open triangles: excitation. Filled circles: inhibition.

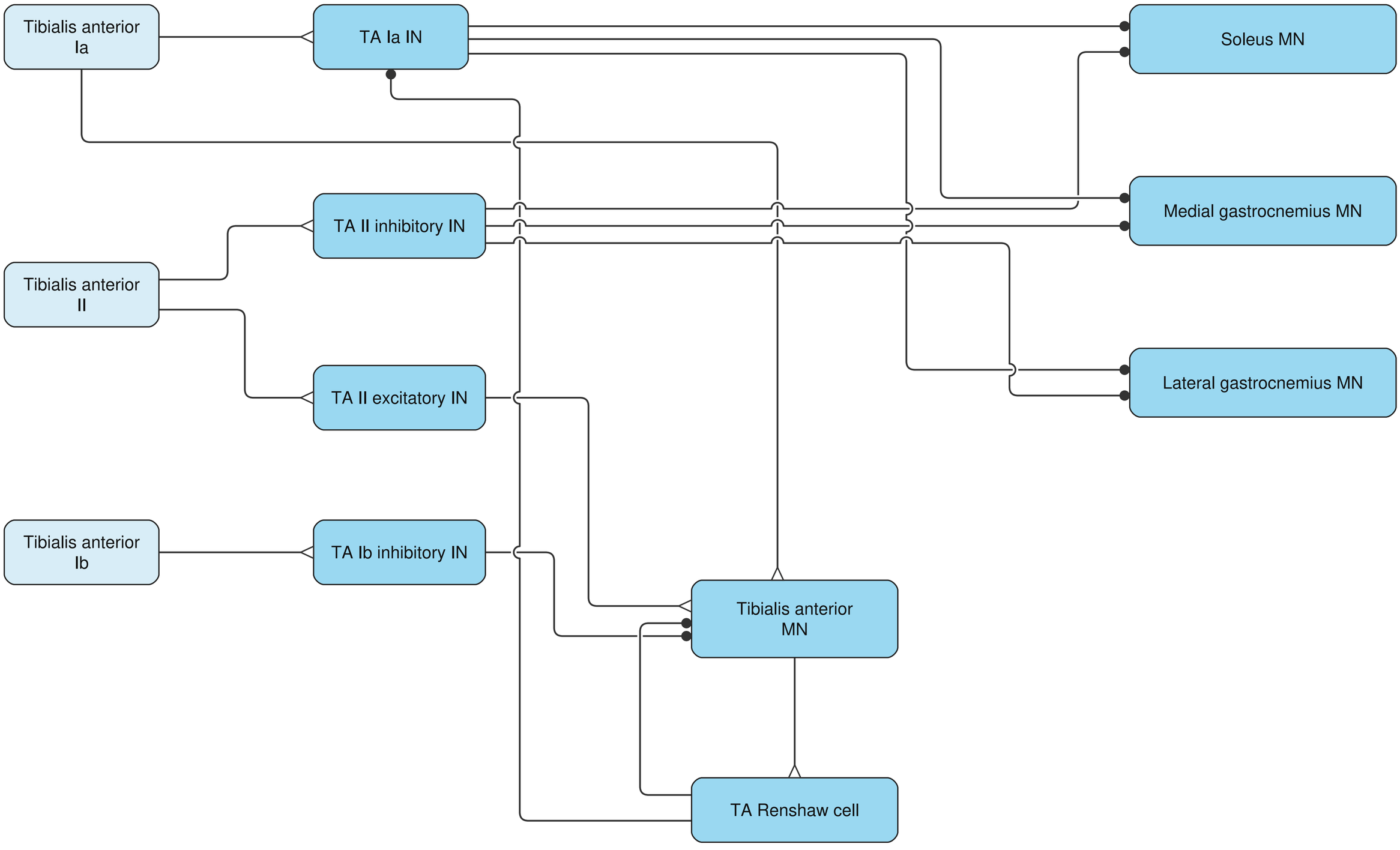
Spindle and tendon symbols show encoded signals; muscle→spindle→AD→Ia/II and muscle→force sensor→AD→Ib are explicit.

MN→AD→muscle carries activation. Each muscle retains its own MN, Ia, II, Ib, interneurons, and Renshaw cell.

Gray dashed PF→MN routes have effective g = 0. Central afferent→RG/PF projections are detailed in the preceding RG/PF sheets.

RF: hip flexion + knee extension. SM: hip extension + knee flexion. Both PF inputs terminate on the same muscle MN.

4 · One muscle’s local motif can inhibit several antagonist MNs



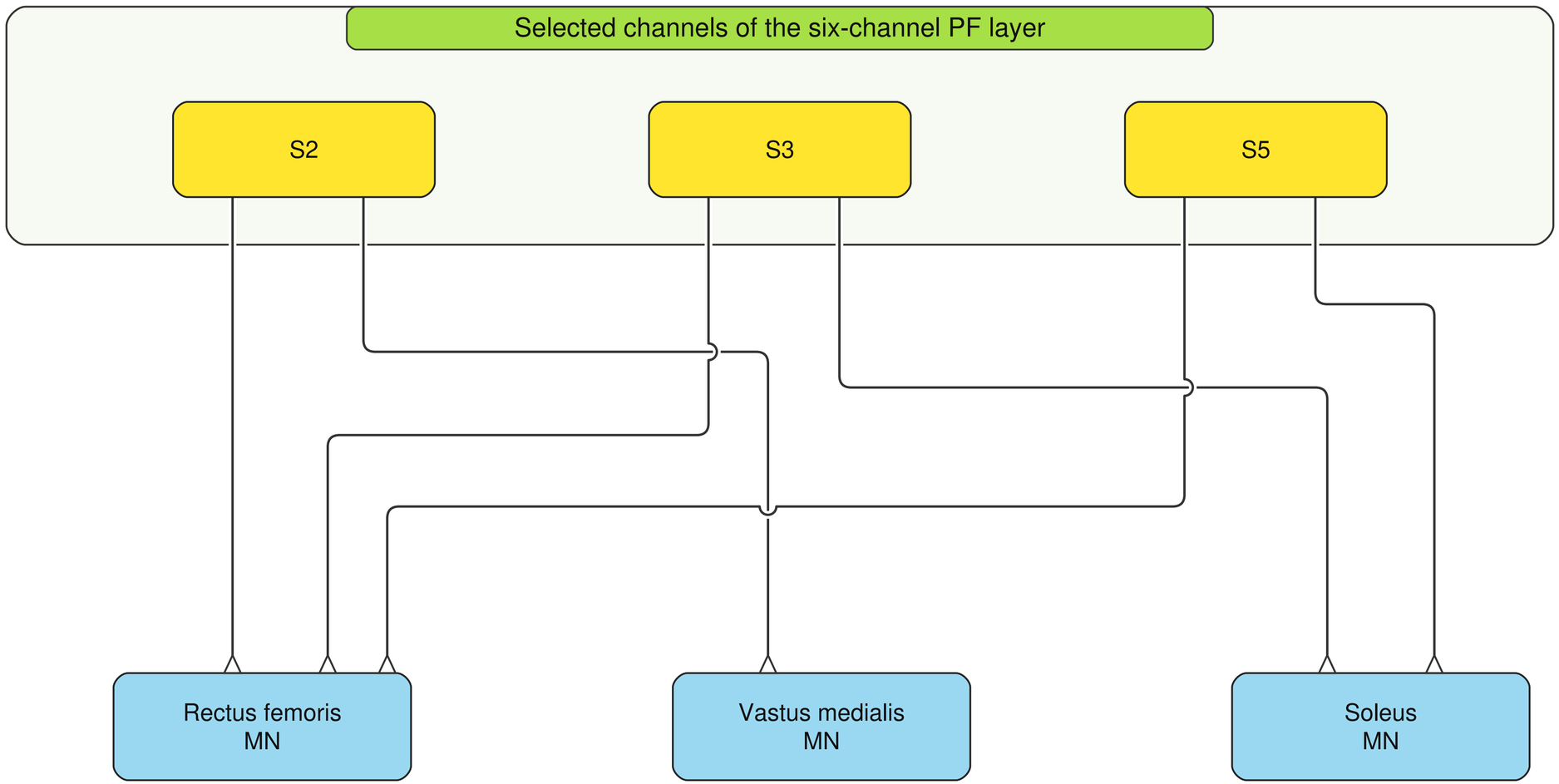
A separate inhibitory synapse reaches each antagonist MN; the three targets are not one pooled muscle.

The same TA Ia neuron also excites its own MN. II uses separate excitatory and inhibitory interneurons.

PF gating, central feedback, other antagonist muscles, and the wider Renshaw network are outside this focused view.

TA = tibialis anterior. All lines on this page are nonzero saved w2lvar connections. No muscle or MN is duplicated.

5 · Active syn6 PF channels converge and fan out across muscles



All three PF inputs to rectus femoris are shown. The other two MNs also receive inputs from channels outside this view.

Rectus femoris integrates three separate PF synapses; S2 also branches to vastus medialis.

S3 and S5 also branch to soleus. These channels encode muscle synergies, not named hip/knee half-centers.

S2 → rectus femoris: g = 0.007152 µS; S2 → vastus medialis: g = 0.01747 µS.

Each outgoing synapse has its own weight. Other channel inputs and local reflex circuits are omitted here.

All displayed connections are excitatory and nonzero. Exact conductances and thresholds accompany the editable drawing.